INSTRUCTOR LESSON PLAN · DAY 1

# NVD Driving: Classroom, Written & Familiarization

AN/PVS-14 · M1075A1 PLS · IAW AR 600-55, TC 21-305-2, AR 385-10, TC 21-60

Date: _____
Instructor: _____
Students: _____

> **Training aid.** Verify all content against the current AR 600-55, TC 21-305-2, AR 385-10, TC 21-60, the AN/PVS-14 and M1075A1 operator TMs, and your unit SOP. Unit SOP and the publications take precedence over anything in this packet.

## Terminal learning objective

**Action:** Explain the requirements, capabilities, limitations, and safe-use procedures for driving the M1075A1 PLS with the AN/PVS-14, and perform PVS-14 PMCS, mounting, and focusing.
**Condition:** In a classroom and a darkened familiarization area, given an AN/PVS-14 with helmet or head mount, batteries, training videos, and this packet.
**Standard:** Score at least _____% on the written exam and receive a GO at each familiarization station.

## Schedule (about 6 hours; adjust times to your day)

| Time | Length | Block | Notes |
|---|---|---|---|
| ____ | 15 min | Opening, safety brief, objectives, roll call | Explain today's GO/NO-GO standard. Confirm every student holds a valid M1075A1 license. |
| ____ | 30 min | Video 3: NVD Safety, plus discussion | Use the discussion questions below. |
| ____ | 40 min | Video 1: NVG Driving Do's & Don'ts, plus discussion | Pause after each DO/DON'T and ask "why?" before playing on. |
| ____ | 10 min | Break | |
| ____ | 35 min | Video 2: Ground Guiding at Night, plus discussion | Have the class demonstrate a few TC 21-60 signals. |
| ____ | 40 min | Written exam | Closed book. Collect all copies. |
| ____ | 20 min | Exam review | Go over every missed question. Retest per SOP. |
| ____ | 45 min | Lunch | Set up stations; darken the room or wait for dusk. |
| ____ | 2 h | Familiarization stations (6 × about 15 min, rotate) | Groups of 2 to 4. Mark GO/NO-GO on the roster. |
| ____ | 15 min | After-action review, accountability | Inventory every device and battery before release. |

## Materials

- Projector, speakers, the 3 videos downloaded in advance (no internet needed)
- AN/PVS-14 with mounts: 1 per 2 students (more if possible)
- AA batteries: 2 per device, plus spares
- Lens paper; daylight training covers if training in daylight
- Focus target (eye chart or printed text) on a wall 20+ ft away
- 6 to 10 traffic cones and a measuring tape
- Flashlights with filtered lenses per SOP; 1 unfiltered white light for the demo
- IR chemlights or IR-filtered wands
- Large-print signs (for the field-of-view station)
- Written exams, answer key, pencils
- Station cards and the GO/NO-GO roster (this packet)
- Sensitive-item hand receipt / sign-out sheet

## Discussion questions

### Video 3: NVD Safety

1. Name three limits of the PVS-14 that cannot be fixed by focusing better.
2. Why do we keep the unaided eye dark-adapted while wearing a monocular?
3. Your tube fails at 25 mph on a dirt road. Talk through your actions, in order.
4. What changes in your plan when illumination is low or there is dust or fog?

### Video 1: Do's & Don'ts

1. What must be on your license record before you drive on NVDs? (AR 600-55, DA Form 5984-E)
2. Why focus the diopter before the objective?
3. What does "overdriving your goggles" mean? How do you prevent it?
4. What is the A-driver's job during NVD driving, and why the right side especially?
5. Why is following distance harder to judge through a monocular?

### Video 2: Ground Guiding at Night

1. Where does the ground guide stand when backing the PLS, and why?
2. Who can give the stop signal? Who gives directions?
3. Why can IR signals be dangerous in a mixed crew (some on NVDs, some not)?
4. What does the driver do the moment sight of the guide is lost?

## Familiarization stations

> Run in a darkened room or after dusk. If training in daylight, use the daylight training cover per the TM. Every station has one instructor or assistant instructor. Account for every device and battery at the end of each rotation.

### Station 1 · PMCS and mounting

About 15 min · Ref: AN/PVS-14 operator TM

1. Inspect the housing, lenses, eyecup, and mount for cracks, scratches, dirt, or missing parts.
2. Install the battery with correct polarity. Confirm the student has a spare.
3. Mount the device on the helmet or head mount. Lock it in place and check it is stable when the head moves.
4. Turn on per the TM. Check for the low-battery indicator and any image defects. Turn off.

**GO when:** the student finds a planted defect or names every check, mounts the device securely, and powers it on and off correctly.

### Station 2 · Focusing

About 15 min · Ref: TC 21-305-2, operator TM

1. Adjust the eyepiece diopter until the image noise (scintillation) is sharp.
2. Adjust the objective focus on the target at 20+ ft until it is sharp.
3. Read the target aloud. The instructor checks it.
4. Defocus both; the student refocuses from scratch.

**GO when:** the student focuses in the correct order (diopter, then objective) and reads the target twice.

### Station 3 · Field of view and scanning

About 15 min

1. The student looks straight ahead through the PVS-14, unaided eye covered.
2. Assistants hold signs at about 30°, 60°, and 90° to the left and right.
3. The student reports what they can read without turning the head, then scans and reads every sign.

**GO when:** the student scans left, center, and right and reads every sign.
**Teaching point:** the 40° field of view hides most of the scene until you turn your head.

## Station 4 · Depth perception

About 15 min

1. Set cones at unmarked distances (for example 15, 30, 50, and 80 ft).
2. Through the PVS-14, the student estimates the distance to each cone. Record the estimates.
3. The student paces or tapes the real distances and compares.

**GO when:** the student completes the estimates and explains why intervals must increase on NVDs.

## Station 5 · Light discipline and bright sources

About 15 min · Instructor controls the white light at all times

1. Student on the PVS-14 in the dark room. The instructor briefly turns on a filtered light, then an unfiltered white light, off to the side. Never aim it at the student's eyes or the tube.
2. The student describes the blooming and loss of detail, then looks away and reports recovery.
3. Show IR chemlights: invisible to the naked eye, bright through the tube.

**GO when:** the student explains blooming, white-light rules in the cab, and who can and cannot see IR.

## Station 6 · Ground-guide signals

About 15 min · Ref: TC 21-60, unit SOP

1. In pairs: one guide, one "driver". Rehearse the briefed signals: come back, turn, slow, stop.
2. First with filtered flashlights (driver unaided), then with IR chemlights or wands (driver on the PVS-14).
3. The guide breaks line of sight; the "driver" calls STOP.

**GO when:** signals are large, slow, and correct, the guide stays on the driver's side, and the driver stops when sight is lost.

## GO / NO-GO roster

| # | Name / Rank | License on M1075A1 ✓ | Exam % | S1 | S2 | S3 | S4 | S5 | S6 | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | |
| 2 | | | | | | | | | | |
| 3 | | | | | | | | | | |
| 4 | | | | | | | | | | |
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| 11 | | | | | | | | | | |
| 12 | | | | | | | | | | |
| 13 | | | | | | | | | | |
| 14 | | | | | | | | | | |
| 15 | | | | | | | | | | |
| 16 | | | | | | | | | | |

Mark each station G (GO) or NG (NO-GO). Retrain and retest NO-GOs before the night practical. Record completed training on the operator's license record per AR 600-55 and unit SOP.

Instructor signature: _____ Date: _____